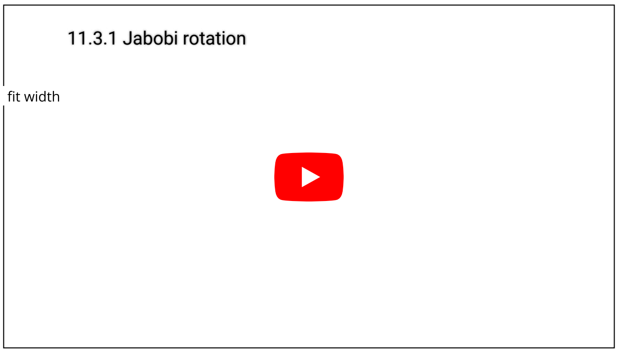
11.3.1 Jacobi rotation ¶



Given a symmetric 2×2 matrix A, with

$$A = \begin{pmatrix} \alpha_{0,0} & \alpha_{0,1} \\ \alpha_{1,0} & \alpha_{1,1} \end{pmatrix}$$

There exists a rotation,

$$J = \begin{pmatrix} \gamma & -\sigma \\ \sigma & \gamma \end{pmatrix},$$

where $\gamma = \cos(\theta)$ and $\sigma = \sin(\theta)$ for some angle θ, such that

$$J^T A J = \begin{pmatrix} \gamma & \sigma \\ -\sigma & \gamma \end{pmatrix} \begin{pmatrix} \alpha_{0,0} & \alpha_{0,1} \\ \alpha_{1,0} & \alpha_{1,1} \end{pmatrix} \begin{pmatrix} \gamma & -\sigma \\ \sigma & \gamma \end{pmatrix} = \begin{pmatrix} \lambda_0 & 0 \\ 0 & \lambda_1 \end{pmatrix} = \Lambda.$$

We recognize that

$$A = J \Lambda J^T$$

is the Spectral Decomposition of A. The columns of J are eigenvectors of length one and the diagonal elements of Λ are the eigenvalues.

Ponder This 11.3.1.1. Give a geometric argument that Jacobi rotations exist.

▼ Hint

For this exercise, you need to remember a few things:

- How is a linear transformation, L, translated into the matrix A that represents it, $Ax = L(x)$?
- What do we know about the orthogonality of eigenvectors of a symmetric matrix?
- If A is not already diagonal, how can the eigenvectors be chosen so that they have unit length, first one lies in Quadrant I of the plane, and the other one lies in Quadrant II?
- Draw a picture and deduce what the angle θ must be.

It is important to note that to determine J we do not need to compute θ. We merely need to find one eigenvector of the 2×2 matrix from which we can then compute an eigenvector that is orthogonal. These become the columns of J. So, the strategy is to

- Form the characteristic polynomial

  $$\begin{array}{rcl} \det(\lambda I - A) & = & (\lambda - \alpha_{0,0})(\lambda - \alpha_{1,1}) - \alpha_{1,0}^2 \\ & = & \lambda^2 - (\alpha_{0,0} + \alpha_{1,1})\lambda + (\alpha_{0,0}\alpha_{1,1} - \alpha_{1,0}^2), \end{array}$$

  and solve for its roots, which give us the eigenvalues of A. Remember to us the stable formula for computing the roots of a second degree polynomial, discussed in Subsection 9.4.1.
- Find an eigenvector associated with one of the eigenvalues, scaling it to have unit length and to lie in either Quadrant I or Quadrant II. This means that the eigenvector has the form

  $$\begin{pmatrix} \gamma \\ \sigma \end{pmatrix}$$

  if it lies in Quadrant I or

  $$\begin{pmatrix} -\sigma \\ \gamma \end{pmatrix}$$

  if it lies in Quadrant II.

This gives us the γ and σ that define the Jacobi rotation.

Homework 11.3.1.2. With Matlab, use the `eig` function to explore the eigenvalues and eigenvectors of various symmetric matrices:

```
[ Q, Lambda ] = eig( A )
```

Try, for example

```
A = [
-1  2
 2  3
]

A = [
 2  -1
-1 -2
]
```

How does the matrix Q relate to a Jacobi rotation? How would Q need to be altered for it to be a Jacobi rotation?

▾ Solution

```
>>  A = [
 -1  2
  2  3
]

A =

    -1     2
     2     3

>> [ Q, Lambda ] = eig( A )

Q =

   -0.9239    0.3827
    0.3827    0.9239


Lambda =

   -1.8284         0
         0    3.8284
```

We notice that the columns of Q need to be swapped for it to become a Jacobi rotation:

$$J = \begin{pmatrix} 0.9239 & 0.3827 \\ -0.3827 & 0.9239 \end{pmatrix} = \begin{pmatrix} 0.3827 & -0.9239 \\ 0.9239 & 0.3827 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = Q \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

```
>> A = [
 2  -1
-1 -2
]

A =

     2    -1
    -1    -2

>> [ Q, Lambda ] = eig( A )

Q =

   -0.2298   -0.9732
   -0.9732    0.2298


Lambda =

   -2.2361         0
         0    2.2361
```

We notice that the columns of Q need to be swapped for it to become a Jacobi rotation. If we follow our "recipe", we also need to negate each column:

$$J = \begin{pmatrix} 0.9732 & -0.2298 \\ 0.2298 & 0.9732 \end{pmatrix}.$$

These solutions are not unique. Another way of creating a Jacobi rotation is to, for example, scale the first column so that the diagonal elements have the same sign. Indeed, perhaps that is the easier thing to do:

$$Q = \begin{pmatrix} -0.9239 & 0.3827 \\ 0.3827 & 0.9239 \end{pmatrix} \quad \longrightarrow \quad J = \begin{pmatrix} 0.9239 & 0.3827 \\ -0.3827 & 0.9239 \end{pmatrix}.$$